Experiment 10

Q1 Answer following question in typed form and paste it in the text box provided for submission: **bold text***

1 Explain input and output characteristics for f_oneway function from stats library

Input and Output Characteristics of f_oneway() Function (stats library)

The f_oneway() function from the scipy.stats library is used to perform a One-Way ANOVA (Analysis of Variance) test. It checks whether the means of two or more groups are significantly different.

Input Characteristics

The f_oneway() function takes two or more sample groups as input.

Inputs include: **bold text** Two or more arrays/lists of data values Each array represents one group or sample. Numeric data The values in each group should be numerical. Independent samples The samples should be independent from each other. Same measurement type All groups should represent the same variable measured under different conditions.

Example:

```
from scipy.stats import f_oneway

group1 = [10, 12, 11, 13]
group2 = [20, 21, 19, 22]
group3 = [15, 16, 14, 17]

f_oneway(group1, group2, group3)
```

```
F_onewayResult(statistic=np.float64(48.79999999999991),
pvalue=np.float64(1.4763314494112902e-05))
```

Output Characteristics

The f_oneway() function returns two values:

F-statistic Measures the ratio of variance between groups to variance within groups. A higher F-value indicates greater difference between group means. p-value Shows the probability that the observed difference occurred by chance. Used to make the decision about the null hypothesis.

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Interpretation If p-value ≤ 0.05 → Reject the null hypothesis → Group means are significantly different. If p-value > 0.05 → Accept the null hypothesis → No significant difference between group means.

2 Explain results of sm.stats.anova_lm fucntion. **bold text**

Results of sm.stats.anova_lm() Function

The sm.stats.anova_lm() function from the statsmodels (sm) library is used to perform ANOVA (Analysis of Variance) on a fitted linear model. It helps determine whether independent variables significantly affect the dependent variable.

Output of anova_lm() Function

The function returns a table with several columns that summarize the ANOVA results.

Column	Meaning
df	Degrees of Freedom for each factor or variable in the model.
sum_sq	Sum of Squares – measures variation explained by each factor.
mean_sq	Mean Square – calculated as `sum_sq / df`.
F	F-statistic – ratio used to test whether the variable significantly affects the outcome.
PR(>F)	p-value – probability of observing the result if the null hypothesis is true.

Interpretation of Results Degrees of Freedom (df) Shows the number of independent values used to estimate the variance. **Sum of Squares (sum_sq)** Represents how much variation in the dependent variable is explained by each independent variable. **Mean Square (mean_sq)** Average variation for each factor. **F-statistic** Used to determine whether the variation explained by a factor is significant. **p-value (PR(>F))** Helps decide whether to reject the null hypothesis. If p-value ≤ 0.05 → Reject null hypothesis → Variable has significant effect. If p-value > 0.05 → Accept null hypothesis → Variable has no significant effect.

Example Output

Source	df	sum_sq	mean_sq	F	PR(>F)
Factor	2	45.6	22.8	5.12	0.01
Residual	27	120.4	4.46		

Interpretation: Since p-value = 0.01 < 0.05, the factor has a significant effect on the dependent variable.

3 What role ANOVA plays in data analytics? **bold text**

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Role of ANOVA in Data Analytics

ANOVA (Analysis of Variance) is a statistical technique used in data analytics to compare the means of two or more groups and determine whether the differences between them are statistically significant.

Roles of ANOVA Compare Multiple Groups **bold text** ANOVA helps analysts compare three or more groups at the same time to see if **their means are different.** Identify Significant Factors It helps determine whether an independent variable significantly affects a **dependent variable.** Reduce Error in Analysis Instead of performing many t-tests, ANOVA analyzes all groups together, reducing the chance of statistical error. **Support Decision Making** Businesses and researchers use ANOVA to identify which factor or treatment gives better results. **Used in Experiments and Research** It is commonly used in A/B testing, product testing, medical research, marketing analysis, and machine learning experiments.

Example

A company wants to know whether three different advertising strategies lead to different sales results. ANOVA can test if the average sales of the three strategies are significantly different.

4 When to use one way ANOVA? Explain with real world case. **bold text**

When to Use One-Way ANOVA

One-Way ANOVA (Analysis of Variance) is used when we want to compare the means of three or more independent groups based on one factor (independent variable) to see if there is a significant difference between them.

Conditions to Use One-Way ANOVA 1 There is one independent variable (factor). 2 There are three or more groups or categories. 3 The dependent variable is numerical (continuous). 4 The samples are independent of each other.

Real-World Example

Case: Teaching Methods and Student Performance

A school wants to check whether different teaching methods affect student exam scores.

Independent Variable: Teaching method Method A Method B Method C **Dependent Variable: Student exam score (numeric)**

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The school collects exam scores from students taught using the three different methods.

Using One-Way ANOVA, the school can test whether the average scores of the three groups are significantly different.

Interpretation **bold text** If p-value ≤ 0.05 → There is a significant difference in exam scores between teaching methods. If p-value > 0.05 → There is no significant difference between the methods.

5 When to use 2 way ANOVA ? Explain with real world case. **bold text**

When to Use Two-Way ANOVA

Two-Way ANOVA is used when we want to study the effect of two independent variables (factors) on one dependent variable and also check if there is interaction between the two factors.

Conditions to Use Two-Way ANOVA There are two independent variables (factors). There is one dependent variable (numeric/continuous). Data is collected from two or more groups for each factor. Samples are independent.

Real-World Example

Case: Study Hours and Teaching Method on Exam Scores

A college wants to check how teaching method and study hours affect student exam scores.

Independent Variable 1: Teaching Method Online Offline Independent Variable 2: Study Hours 2 hours 4 hours Dependent Variable: Exam Score (numeric)

Using Two-Way ANOVA, the college can determine:

Whether teaching method affects exam scores. Whether study hours affect exam scores. Whether there is an interaction between teaching method and study hours.

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```
**Conclusion**
Two-Way ANOVA is useful in data analysis
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Conclusion Two-Way ANOVA is useful in data analysis when we want to study the impact of two different factors on a single outcome and also understand how these factors interact with each other. It helps researchers and analysts make better decisions based on multiple variables at the same time.

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